

Finite Distribution Modules, Spectral Jets, and Cyclotomic Trace Identities

Uniform rank proofs, explicit normal forms, exceptional zeros,
and an exact audit of a Gaussian weight-five conjecture

Research contribution prepared for the ProveIt project
OpenAI-assisted mathematical research; supplied for independent review

9 October 2026

Abstract

The ProveIt polylogarithm manuscript records finite experiments suggesting that rational distribution relations leave $\varphi(q) - 1$ coordinates at level q , and $\varphi(q)/2 - 1$ after suitable reflection relations. We give a uniform proof for an explicitly defined presentation, simultaneously for all independent prime weights and every finite spectral-jet order. Over a character splitting field, the universal weighted quotient is a free polynomial module of rank $\varphi(q)$. Its conductor-indexed basis remains valid at resonant weights, whereas the basis consisting of primitive-grid points can fail. We factor the determinant of that change of coordinates, compute its local jet defects, and give rational primitive-grid normal forms with exact support, sign, and denominator information.

The resulting calculus yields all-orders Hurwitz–Stieltjes reduction identities, including the pole correction for underived Stieltjes constants. Applying the same module to roots-of-unity polylogarithms gives character trace identities and exact orders of vanishing at spectral order one; for example, the primitive level-30 trace has second order derivative $-2 \log 2 \log 3 \log 5$. Separately, a rational certificate shows that the manuscript’s S_4 conjecture is not a consequence of a specified 96-row weight-five double-shuffle presentation, without asserting that the conjecture is false. An audit also corrects the status of the cubic log-gamma moment using the published work of Bailey–Borwein–Borwein. Complete proofs, exact regression programs, replayable certificates, numerical diagnostics, and integration notes accompany the article.

Scope of claims. All rank statements concern explicitly presented formal modules. They do not assert linear independence of transcendental values. No proof-assistant verification or global priority claim is made. Classical antecedents and the remaining conjecture are identified explicitly.

Contents

1	Research target, provenance, and mathematical context	3
1.1	The question resolved here	3
1.2	Classical antecedents and the contribution’s status	3
1.3	Notation and endpoint convention	3
2	The universal weighted distribution module	4
2.1	Proof: identify the complete relation module	5
2.2	A replayable symbolic reduction certificate	6
3	Anchoring, reflection, and uniform jet ranks	6
3.1	The precise replacement for the finite rank observation	6
3.2	All finite jets by base change	7
4	The primitive-grid determinant and its exceptional locus	7
4.1	Local jet defects of the primitive-grid map	8
5	Explicit primitive-grid reduction coefficients	9
5.1	A finite formula, including negative spectral parameters	10
5.2	Denominator certificates and construction cost	12
6	All-orders Hurwitz–Stieltjes identities	12
7	Cyclotomic polylogarithm traces	14
7.1	Exact nonprincipal vanishing orders	15
7.2	The principal trace and its pole cancellation	15
7.3	Concrete exact identities	16
7.4	A stable independent formula for numerical checks	17
8	The Gaussian S_4 conjecture: an exact obstruction	17
8.1	A fully specified finite law vocabulary	18
8.2	Fresh numerical evidence, with its exact limitation	19
9	Corrections and clarifications for the manuscript	20
9.1	Replace the distribution-rank experiment by a defined theorem	20
9.2	The cubic log-gamma moment already has a named-constant formula	20
9.3	At mixed Gaussian points the relevant involution swaps arguments	20
9.4	Keep three meanings of a certificate distinct	21
10	Reproducible computations and integration	21
10.1	Exact checks actually executed	21
10.2	Numerical diagnostics actually executed	21
10.3	Files and reproduction	22
11	Further research questions	22
11.1	Complete the proof of the weight-five conjecture	22
11.2	Integral and modular versions of the weighted module	22
11.3	Multivariate jets and intersection multiplicities	22
11.4	Sparse algorithms and certified numerical evaluation	23
11.5	Trace identities at greater depth	23
11.6	Arithmetic meaning of the trace-jet constants	23
11.7	Log-gamma moments beyond the known Tornheim formula	23
11.8	A proof-assistant development	23
A	The known cubic log-gamma formula in Tornheim notation	24
B	Provenance, claim boundaries, and artifact checks	25

1 Research target, provenance, and mathematical context

1.1 The question resolved here

The starting point is Chapters 4, 7, 8, and 10 of the ProveIt manuscript *Polylogarithms and their Arithmetic Bridges* [1]. Chapter 8, Observation `tower:thm:rank`, reports exact finite experiments with $3 \leq q \leq 30$ and three parameter-derivative orders:

$$(q - 1) - \text{rank}(\text{distribution rows}) = \varphi(q) - 1. \quad (1.1)$$

Chapter 7 gives the analogous count $\varphi(q)/2 - 1$ after distribution and first-Stieltjes reflection. The manuscript correctly distinguishes these formal calculations from numerical independence and explicitly asks for a proof uniform in the denominator and derivative order.

The historical matrices are not included in the cited manuscript snapshot. Consequently we do not claim to reconstruct their exact row ordering or to certify unseen computations. Instead, we define the canonical complete divisor-row presentation from the stated distribution law, prove its rank for every q , and give code generating every row. This resolves the uniform mathematical problem underlying the observation and supplies a precise replacement for its experimental formulation.

The stronger result is useful: the rank does not depend on the multiplicative weights at all. Apparent failures at exceptional spectral parameters are failures of a chosen coordinate basis, not jumps in the rank of the complete relation module. A determinant factorization makes this distinction quantitative.

1.2 Classical antecedents and the contribution's status

Universal distributions are classical. In particular, the ordinary specialization of our module, with all prime weights equal to one, is a finite-level universal ordinary distribution. Ouyang's Proposition 3.1 records its integral freeness and rank $\varphi(q)$, citing the work of Kubert and Anderson [2, 3]. We do not present that specialization as a new theorem.

The contribution developed here is an explicit polynomial-weight presentation with conductor normal forms, its determinant and jet specializations, and the resulting integration-ready proof of the manuscript's rank problem. The polylogarithm trace formulas derive from classical distribution and Gauss-sum identities; their concrete order-derivative evaluations are supplied as consequences, without a claim that each has not appeared elsewhere. The exact S_4 row-space obstruction concerns a newly specified finite presentation, rather than all possible multiple-polylogarithm identities.

Standard analytic conventions for Hurwitz zeta, polylogarithms, and Dirichlet L -functions are as in the corresponding DLMF sections [4, 5, 6]. Every additional algebraic rank assertion used here is proved below rather than inferred from a numerical relation search.

1.3 Notation and endpoint convention

For $q \geq 2$, let

$$X_q = (q^{-1}\mathbb{Z})/\mathbb{Z}, \quad U_m = (\mathbb{Z}/m\mathbb{Z})^\times, \quad P(q) = \{p : p \text{ prime, } p \mid q\}.$$

We use U_1 as a one-element group and put $\varphi(1) = 1$. The point $0 \in X_q$ is a *formal* point. Under Hurwitz evaluation it means the endpoint 1, never the undefined expression $\zeta(s, 0)$:

$$e_{a/m} \mapsto \zeta(s, a/m) \quad (1 \leq a < m), \quad e_0 \mapsto \zeta(s, 1).$$

Under cyclotomic evaluation it means $\text{Li}_s(1) = \zeta(s)$, initially away from the pole. This convention is essential when a distribution fiber passes through zero.

Every character of U_q has a unique primitive inducing character of conductor $f \mid q$. A symbol χ will denote that primitive character and, where specified, its inflation to U_m for $f \mid m \mid q$. The principal character has conductor one. Character values at an integer prime to f are roots of unity; primes dividing f are treated separately, so no ambiguous evaluation at a nonunit is needed.

2 The universal weighted distribution module

Fix a characteristic-zero field K containing all values of characters of U_q , and put

$$R = K[A_p : p \in P(q)], \quad \mathcal{V}_q = \bigoplus_{x \in X_q} R e_x.$$

The A_p are algebraically independent indeterminates. For each $p \mid q$ and $x \in (p/q)\mathbb{Z}/\mathbb{Z}$, define a prime distribution row

$$r_{p,x} = \sum_{\substack{y \in X_q \\ py=x}} e_y - A_p e_x. \quad (2.1)$$

Let $\mathcal{R}_q$ be their R -span and let $\mathcal{D}_q = \mathcal{V}_q/\mathcal{R}_q$.

For $d \mid q$ write $A_d = \prod_{p \mid d} A_p^{v_p(d)}$. The corresponding divisor row is

$$r_{d,x} = \sum_{dy=x} e_y - A_d e_x, \quad x \in (d/q)\mathbb{Z}/\mathbb{Z}. \quad (2.2)$$

The multiplicativity of A_d is part of the definition; arbitrary unrelated weights on composite divisors would be a different problem.

Lemma 2.1 (Prime rows suffice). *Every row (2.2) lies in $\mathcal{R}_q$.*

Proof. If $d = ab \mid q$, then for $x \in (d/q)\mathbb{Z}/\mathbb{Z}$ every solution of $ay = x$ lies in $(b/q)\mathbb{Z}/\mathbb{Z}$, and

$$r_{ab,x} = \sum_{ay=x} r_{b,y} + A_b r_{a,x}.$$

Each z with $abz = x$ occurs exactly once in the double sum, and the intermediate e_y terms cancel. Induction on the number of prime factors of d , counted with multiplicity, proves the assertion. No division by any weight is used. $\square$

For $f = \text{cond}(\chi)$ and $f \mid m \mid q$, set

$$v_{m,\chi} = \sum_{a \in U_m} \chi(a) e_{a/m}, \quad v_{1,1} = e_0. \quad (2.3)$$

Define the polynomial

$$E_{m,\chi} = \prod_{p \mid f} A_p^{v_p(m) - v_p(f)} \prod_{\substack{p \mid m \\ p \nmid f}} A_p^{v_p(m) - 1} (A_p - \chi(p)). \quad (2.4)$$

In particular $E_{f,\chi} = 1$.

Theorem 2.2 (Universal conductor normal form). *The quotient $\mathcal{D}_q$ is a free R -module of rank $\varphi(q)$. One basis consists of*

$$b_\chi = [v_{f,\chi}], \quad f = \text{cond}(\chi), \quad \chi \bmod q.$$

In this basis,

$$[v_{m,\chi}] = E_{m,\chi} b_\chi \quad (f \mid m \mid q). \quad (2.5)$$

The relation module is free of rank $q - \varphi(q)$ and is a direct summand of $\mathcal{V}_q$. All these statements survive arbitrary specialization of the weights into any K -algebra, including $A_p = 0$ and $A_p = 1$.

2.1 Proof: identify the complete relation module

It is not enough to exhibit character identities satisfied by evaluated special functions: that would give only one direction of a rank argument. We therefore identify the image of the entire relation map.

The points of X_q split by exact denominator. Hence

$$\mathcal{V}_q = \bigoplus_{m \mid q} R[U_m]. \quad (2.6)$$

The reduction map $U_q \rightarrow U_m$ is surjective. Character orthogonality therefore gives an invertible Fourier change of basis on each summand. A character of conductor f occurs exactly once in the summand indexed by m when $f \mid m$, and does not occur otherwise. Thus the χ block of (2.6) has basis

$$v_{m,\chi}, \quad f \mid m \mid q.$$

Fix $p \mid q$. The source labels x of the rows $r_{p,x}$ themselves form a grid of level q/p . Fourier-transforming separately on its exact-denominator strata gives all transformed rows, not a selected subset of them. For $f \mid m$ and $pm \mid q$, the transformed row is

$$v_{pm,\chi} - \lambda_{p,m,\chi} v_{m,\chi}, \quad \lambda_{p,m,\chi} = \begin{cases} A_p, & p \mid m, \\ A_p - \chi(p), & p \nmid m. \end{cases} \quad (2.7)$$

To check this directly, weight the row at $x = a/m$ by $\chi(a)$ and sum over U_m . If $p \mid m$, all p preimages have exact denominator pm . If $p \nmid m$, one preimage has denominator m : it is b/m with $pb \equiv a \pmod{m}$, and its coefficient is $\chi(a) = \chi(p)\chi(b)$. All other preimages have denominator pm . This proves (2.7). Since the Fourier transforms of the source and target are invertible, these edges span precisely the original relation module in character coordinates.

Starting at $m = f$, multiply the edge factors while adjoining prime factors of m/f . A prime already dividing f always contributes A_p . A prime not dividing f contributes $A_p - \chi(p)$ on its first appearance and A_p on every later appearance. The resulting product is exactly (2.4), independently of the order of adjoining primes. Therefore all $v_{m,\chi}$ reduce to $E_{m,\chi} v_{f,\chi}$.

There are no additional relations on $v_{f,\chi}$. Indeed, the map from the free χ block to R defined by

$$v_{m,\chi} \mapsto E_{m,\chi}$$

annihilates every edge and sends $v_{f,\chi}$ to one. It induces an inverse to $R \rightarrow \mathcal{D}_{q,\chi}$, $1 \mapsto [v_{f,\chi}]$. Moreover, its kernel has the explicit monic basis

$$v_{m,\chi} - E_{m,\chi} v_{f,\chi}, \quad f \mid m \mid q, \quad m \neq f. \quad (2.8)$$

These vectors are independent by their distinct non-base coordinates. There is one quotient coordinate for each character of U_q , hence $\varphi(q)$ in all, and the complementary kernel has rank $q - \varphi(q)$. This is a split decomposition over the polynomial ring itself, so any base change preserves it. $\square$

Corollary 2.3 (Pointwise polynomial normal form). *For a reduced point a/m with $m \mid q$,*

$$[e_{a/m}] = \frac{1}{\varphi(m)} \sum_{\chi \bmod m} \overline{\chi(a)} E_{m,\chi} b_\chi. \quad (2.9)$$

The sum runs over characters of U_m , identified with their primitive inducing characters and then inflated to U_q .

Proof. Apply inverse character Fourier transformation to (2.3), then use (2.5). $\square$

2.2 A replayable symbolic reduction certificate

Choose a chain $f = m_0 \mid m_1 \mid \cdots \mid m_t = m$ in which each ratio is prime. Let λ_j be its edge factor. The exact identity

$$v_{m,\chi} - E_{m,\chi} v_{f,\chi} = \sum_{j=1}^t \left(\prod_{h=j+1}^t \lambda_h \right) (v_{m_j,\chi} - \lambda_j v_{m_{j-1},\chi}) \quad (2.10)$$

is a finite row-combination certificate. Its verification is polynomial cancellation. In particular, a certificate does not become invalid when one or several factors vanish. This is the practical advantage of conductor coordinates over a generic matrix inverse.

3 Anchoring, reflection, and uniform jet ranks

3.1 The precise replacement for the finite rank observation

For actual evaluations, the endpoint value and lower-order terms can be regarded as known. In a homogeneous difference system this amounts to setting the corresponding formal coordinates equal to zero.

Corollary 3.1 (Anchored rank). *At every specialization of the weights, the quotient by the distribution relations and $e_0 = 0$ has dimension $\varphi(q) - 1$. Equivalently, after deleting the endpoint column, the prime or complete divisor-row matrix on the remaining $q - 1$ columns has rank $q - \varphi(q)$.*

Proof. The endpoint is b_1 , the basis vector of conductor one. It is nonzero and independent of every other conductor coordinate. Adding its row increases the full relation rank by one. Eliminating that endpoint column leaves rank $q - \varphi(q)$ on $q - 1$ columns. $\square$

This statement applies, in particular, to the homogeneous highest-order part of the manuscript's $\gamma_1^{(k)}$ distribution equations, where $A_p = p^{k+1}$. It holds for every $q \geq 2$ and $k \geq 1$, not only the finite experimental range. Known inhomogeneous right sides do not alter this rank provided the system is consistent; the analytic evaluations supply consistency.

The reflection operator $J(e_x) = e_{-x}$ preserves the relation module. Since $Jv_{m,\chi} = \chi(-1)v_{m,\chi}$, its action is diagonal on the conductor basis.

Corollary 3.2 (Parity counts). *For $q > 2$, the unanchored even and odd quotients have dimensions $\varphi(q)/2$ each. After anchoring the endpoint, their dimensions are respectively $\varphi(q)/2 - 1$ and $\varphi(q)/2$. Thus imposing known odd reflection differences leaves exactly $\varphi(q)/2 - 1$ formal homogeneous coordinates.*

Proof. The element $-1 \in U_q$ has order two and is not the identity. Exactly half the characters take value 1 there and half take value -1 . The principal character is even, so anchoring removes one even coordinate. For $q = 2$ the separate counts are zero even coordinates after anchoring and zero odd coordinates. $\square$

This proves the canonical formal version of the Chapter 7 count. It does not say that $\varphi(q)/2 - 1$ numerical Stieltjes values are independent: additional identities may impose further relations after evaluation.

3.2 All finite jets by base change

Let F be a field extending K , and consider the truncated ring

$$T_N = F[t]/(t^{N+1}), \quad N \geq 0.$$

Choose arbitrary truncated prime weights $A_p(t) \in T_N$. Of particular interest are

$$A_p(t) = p^{s_0} \exp(t \log p) \pmod{t^{N+1}} \quad (3.1)$$

for Hurwitz zeta, and $p^{1-s_0} \exp(-t \log p)$ for polylogarithms. The coefficient field may be $\mathbb{C}$, or a field containing the displayed constants; no algebraic independence of actual logarithms is assumed.

Theorem 3.3 (Uniform finite-jet ranks). *The truncated distribution quotient is a free T_N -module of rank $\varphi(q)$ for every choice of $A_p(t)$. Its dimension over F is $(N+1)\varphi(q)$, and the full coefficient-expanded relation matrix has rank*

$$(N+1)(q - \varphi(q)). \quad (3.2)$$

After fixing all endpoint coefficients its dimension is $(N+1)(\varphi(q) - 1)$. The anchored even and odd dimensions for $q > 2$ are $(N+1)(\varphi(q)/2 - 1)$ and $(N+1)\varphi(q)/2$.

Proof. Apply the homomorphism $R \rightarrow T_N$ that sends each indeterminate to the chosen truncated weight. The split decomposition proved in Theorem 2.2 remains split under this base change. Expanding in the basis $1, t, \dots, t^N$ multiplies every free rank by $N+1$. Anchoring and reflection act on the same conductor coordinates as before. This argument is stronger than an appeal to a block-triangular matrix with a possibly singular diagonal: it applies at every weight. $\square$

If all lower jets are already fixed and only the highest jet is unknown, the homogeneous matrix is the order-zero distribution matrix. Its anchored residual dimension is therefore $\varphi(q) - 1$. The full stacked system and this one-layer system are different matrices, and (3.2) should not be confused with the one-layer rank.

4 The primitive-grid determinant and its exceptional locus

Conductor coordinates are polynomial and globally valid, but it is often more convenient to use the $\varphi(q)$ primitive-grid points $e_{u/q}$, $u \in U_q$. Fourier-transforming those points gives $v_{q,\chi}$. By Theorem 2.2, the map from their formal span into $\mathcal{D}_q$ is diagonal in character coordinates:

$$v_{q,\chi} \mapsto E_{q,\chi} b_\chi. \quad (4.1)$$

Its determinant in these specified Fourier and conductor bases is $\Delta_q = \prod_{\chi \bmod q} E_{q,\chi}$.

For $p^e \parallel q$, write

$$M_p = q/p^e, \quad h_p = \text{ord}_{M_p}(p),$$

with $h_p = 1$ when $M_p = 1$.

Theorem 4.1 (Determinant factorization). *One has the polynomial identity*

$$\Delta_q(\mathbf{A}) = \prod_{p^e \parallel q} A_p^{\varphi(M_p)(p^{e-1}-1)} (A_p^{h_p} - 1)^{\varphi(M_p)/h_p}. \quad (4.2)$$

Thus primitive-grid coordinates form a basis precisely when all the factors in (4.2) are invertible in a field specialization. Their failure does not change the rank of $\mathcal{D}_q$.

Proof. Fix $p^e \parallel q$ and inspect the p factor of $\prod_{\chi} E_{q,\chi}$. Characters whose conductor is not divisible by p are exactly the characters of U_{M_p} . Their additional factors are $A_p - \chi(p)$. In the character group of U_{M_p} , the values $\chi(p)$ run through every h_p th root of unity with multiplicity $\varphi(M_p)/h_p$. This follows by duality from the fact that p has order h_p in U_{M_p} . Their product is therefore $(A_p^{h_p} - 1)^{\varphi(M_p)/h_p}$.

It remains to count the pure powers of A_p . If $a = v_p(f)$, the exponent contributed by one character is $e - \max(1, a)$. Writing this as a sum of indicators gives

$$\sum_{\chi \bmod q} (e - \max(1, v_p(f_\chi))) = \sum_{j=1}^{e-1} \#\{\chi : v_p(f_\chi) \leq j\}.$$

The count in the last sum is $\varphi(p^j)\varphi(M_p)$, since these are exactly the characters induced from level $p^j M_p$. Hence the exponent is $\varphi(M_p) \sum_{j=1}^{e-1} \varphi(p^j) = \varphi(M_p)(p^{e-1} - 1)$. Multiply over the primes. $\square$

Example 4.2 (Level twelve). Let χ_3 and χ_4 be the nonprincipal quadratic characters of conductors three and four, and let $\chi_{12} = \chi_3 \chi_4$. Their conductor generators are

$$\begin{aligned} b_3 &= e_{1/3} - e_{2/3}, & b_4 &= e_{1/4} - e_{3/4}, \\ b_{12} &= e_{1/12} - e_{5/12} - e_{7/12} + e_{11/12}. \end{aligned}$$

For $r \in U_{12}$, the normal form reads

$$\begin{aligned} [e_{r/12}] &= \frac{1}{4} [A_2(A_2 - 1)(A_3 - 1)e_0 + \chi_3(r)A_2(A_2 + 1)b_3 \\ &\quad + \chi_4(r)(A_3 + 1)b_4 + \chi_{12}(r)b_{12}]. \end{aligned} \quad (4.3)$$

Consequently

$$\Delta_{12} = A_2^2(A_2^2 - 1)(A_3^2 - 1).$$

At $A_2 = A_3 = 1$, the primitive-grid span misses the principal conductor coordinate, even though the full quotient still has rank four.

4.1 Local jet defects of the primitive-grid map

Specialize to $A_p(s) = p^s$, and let $s = s_0 + t$. All exponential factors are nonzero. A factor $p^s - \chi(p)$ vanishes precisely when $p^{s_0} = \chi(p)$, and any such zero is simple because its derivative is $p^{s_0} \log p \neq 0$.

Theorem 4.3 (Local Smith factors and primitive-jet ranks). *Define*

$$\rho_\chi(s_0) = \#\{p \mid q : p \nmid f_\chi, p^{s_0} = \chi(p)\}. \quad (4.4)$$

Over $\mathbb{C}[[t]]$, the diagonal factors of the primitive-grid map are units times $t^{\rho_\chi(s_0)}$. Consequently its rank on order- N truncated jets, as a complex linear map, is

$$\sum_{\chi \bmod q} \max\{N + 1 - \rho_\chi(s_0), 0\}. \quad (4.5)$$

At $s_0 = 0$, this is determined by $\rho_\chi(0) = \#\{p \mid q : p \nmid f_\chi, \chi(p) = 1\}$.

Proof. Factor $E_{q,\chi}(s_0 + t)$ into its simple vanishing factors and a unit. The primitive Fourier matrix is constant and invertible, so these are also the local invariant factors, up to permutation and units. Multiplication by t^r on $\mathbb{C}[t]/(t^{N+1})$ has rank $\max(N + 1 - r, 0)$. Summing over the diagonal coordinates proves the formula. This is the rank of a *coordinate inclusion map*, not the rank of the distribution relation matrix. $\square$

Corollary 4.4 (Multiple-prime resonances are central). *If $s_0 \neq 0$, at most one prime can satisfy $p^{s_0 h_p} = 1$. At a nonzero resonant point for that prime p , the primitive-grid rank defect is $\varphi(M_p)/h_p$, and every affected local factor has order one. The order- N inclusion rank is then $(N + 1)\varphi(q) - \varphi(M_p)/h_p$ for all $N \geq 0$.*

Proof. A resonance forces s_0 to be purely imaginary. Resonances at distinct primes would make $\log p / \log \ell$ rational, because each phase is a rational multiple of 2π . Unique prime factorization rules this out unless $s_0 = 0$. For the one resonant prime, the multiplicity count in Theorem 4.1 gives the number of affected characters. Apply Theorem 4.3. $\square$

Example 4.5 (Two exact jet-rank profiles). At level 12 and $s_0 = 0$, the principal character has $\rho = 2$ and the other three characters have $\rho = 0$. For $N = 0, 1, 2$, the primitive-jet ranks are 3, 6, 10, while the full quotient dimensions are 4, 8, 12. At level 21, the principal character has $\rho = 2$, the character induced from conductor three has $\rho = 1$, and the other ten characters have $\rho = 0$. The inclusion ranks are 10, 21, 33, compared with full quotient dimensions 12, 24, 36. Indeed,

$$\Delta_{21} = (A_3^6 - 1)(A_7 - 1)^2.$$

Exact character-phase certificates for these examples are supplied.

5 Explicit primitive-grid reduction coefficients

Away from the exceptional locus, the conductor normal form can be inverted to express all grid points through the primitive ones. For Hurwitz weights there is a particularly concrete answer that avoids character-field arithmetic.

Let $\mathcal{S}_q$ be the multiplicative semigroup of positive integers whose prime factors divide q , including 1. For $\operatorname{Re} s > 0$, define

$$C_{r,u}(s) = \sum_{\substack{d \in \mathcal{S}_q \\ du \equiv r \pmod{q}}} d^{-s}, \quad 0 \leq r < q, \quad u \in U_q. \quad (5.1)$$

The series is absolutely convergent, since $\sum_{d \in \mathcal{S}_q} d^{-\sigma} = \prod_{p|q} (1 - p^{-\sigma})^{-1}$ for $\sigma > 0$.

Theorem 5.1 (Primitive-grid extension). *For $\operatorname{Re} s > 0$, the $q \times \varphi(q)$ matrix $C(s)$ satisfies every prime and divisor distribution relation with weights p^s , and its primitive rows form the identity matrix. It is the unique extension matrix from primitive-grid data. In particular,*

$$\zeta(s, r/q) = \sum_{u \in U_q} C_{r,u}(s) \zeta(s, u/q) \quad (5.2)$$

with the endpoint convention for $r = 0$, initially for $\operatorname{Re} s > 1$ and then meromorphically wherever the coefficients are defined.

Proof. For a primitive row r , the congruence $du \equiv r \pmod{q}$ forces d to be prime to q , hence $d = 1$. This gives the identity submatrix. For a prime distribution row indexed by $a \equiv 0 \pmod{p}$,

$$\sum_{pr \equiv a \pmod{q}} C_{r,u}(s) = \sum_{\substack{d \in \mathcal{S}_q \\ pdu \equiv a \pmod{q}}} d^{-s} = p^s \sum_{\substack{d' \in \mathcal{S}_q \\ d'u \equiv a \pmod{q}}} (d')^{-s}.$$

In the last step $d' = pd$; conversely every d' in the last sum is divisible by p , because u is a unit and a is divisible by p . The last expression is $p^s C_{a,u}(s)$. Prime sufficiency gives all divisor rows. Theorem 2.2 and the identity submatrix show that these $\varphi(q)$ independent columns exhaust the solution space. Hurwitz zeta satisfies the same distribution equations by regrouping its absolutely convergent defining series, proving (5.2). $\square$

5.1 A finite formula, including negative spectral parameters

For $0 \leq r < q$, let

$$g = \gcd(r, q), \quad m = q/g, \quad a = r/g, \quad P_r = \{p \mid q : p \nmid m\}.$$

When $r = 0$, this means $g = q$, $m = 1$, $a = 0$. For $p \in P_r$, put $h_{p,m} = \operatorname{ord}_m(p)$, with order one modulo one.

Theorem 5.2 (Finite residue formula). *The meromorphic continuation of (5.1) is*

$$C_{r,u}(s) = g^{-s} \frac{\sum_{\substack{0 \leq b_p < h_{p,m} \\ p \in P_r}} \prod_{p \in P_r} p^{-sb_p} \prod_{p \in P_r} p^{b_p u} \prod_{p \in P_r} p^{b_p a/m}}{\prod_{p \in P_r} (1 - p^{-sh_{p,m}})}. \quad (5.3)$$

Empty products have their usual value one. In particular,

$$C_{0,u}(s) = q^{-s} \prod_{p \mid q} (1 - p^{-s})^{-1}, \quad (5.4)$$

independently of u .

Proof. The congruence in (5.1) forces the valuation of d at an unsaturated prime, one dividing m , to be exactly $v_p(g)$. At a saturated prime, one not dividing m , it is at least $v_p(g)$. Thus $d = g \prod_{p \in P_r} p^{c_p}$ with $c_p \geq 0$, and the remaining congruence is $u \prod p^{c_p} \equiv a \pmod{m}$. Write $c_p = b_p + h_{p,m} j_p$ and sum the independent geometric series in j_p . This proves the formula in $\operatorname{Re} s > 0$ and hence its meromorphic continuation. The endpoint has trivial congruence modulo one. $\square$

Formula (5.3), *not* the divergent smooth-number series, is the definition used at negative real s . It has no pole at any nonzero real s , and the continued distribution identities remain valid there. At a zero of a displayed denominator, one should use the polynomial conductor normal form rather than cancel poles numerically. Some individual coefficients may have removable singularities, but Theorem 4.1 gives the criterion for the entire primitive-grid coordinate system.

Corollary 5.3 (Exact support and signs). *Let H_r be the subgroup of U_m generated by the classes of the primes in P_r . For every nonzero real s ,*

$$C_{r,u}(s) \neq 0 \iff au^{-1} \in H_r.$$

The number of supported primitive coordinates is

$$\frac{\varphi(q)}{\varphi(m)} |H_r|. \quad (5.5)$$

All supported coefficients are positive for $s > 0$. For $s < 0$, they all have sign $(-1)^{|P_r|}$.

Proof. The numerator of (5.3) is a sum of positive real terms and is nonempty exactly under the subgroup condition. The reduction map $U_q \rightarrow U_m$ has fibers of cardinality $\varphi(q)/\varphi(m)$, giving (5.5). For $s > 0$ every denominator factor is positive. For $s < 0$ each of the $|P_r|$ denominator factors is negative, while the numerator and g^{-s} are positive. $\square$

Corollary 5.4 (Complete monotonicity and normalization). *For $s > 0$ and $j \geq 0$, put*

$$W_{r,u}^{[j]}(s) = (-1)^j \frac{d^j}{ds^j} C_{r,u}(s) = \sum_{\substack{d \in \mathcal{S}_q \\ du \equiv r \pmod{q}}} \frac{(\log d)^j}{d^s}. \quad (5.6)$$

These quantities are nonnegative. Moreover, $\sum_u C_{r,u}(1) = 1$ for every r , and $0 \leq C_{r,u}(k) \leq 1$ for integer $k \geq 1$.

Proof. The logarithmically weighted semigroup series converges locally uniformly in $\operatorname{Re} s > 0$, so termwise differentiation is valid. The constant grid vector satisfies the distribution equations at $s = 1$, and its primitive entries are all one. Uniqueness of extension proves the row-sum identity. Monotonicity gives $C_{r,u}(k) \leq C_{r,u}(1) \leq 1$. $\square$

Example 5.5 (Two reductions at denominator six). The nontrivial coefficients for $r = 2$ are

$$C_{2,1}(s) = \frac{2^s}{2^{2s} - 1}, \quad C_{2,5}(s) = \frac{1}{2^{2s} - 1}.$$

Thus, meromorphically,

$$\zeta(s, 1/3) = \frac{2^s \zeta(s, 1/6) + \zeta(s, 5/6)}{2^{2s} - 1}, \quad (5.7)$$

$$\zeta(s, 1/2) = \frac{\zeta(s, 1/6) + \zeta(s, 5/6)}{3^s - 1}. \quad (5.8)$$

At $s = 1$, the coefficients in (5.7) are $2/3$ and $1/3$. At $s = -1$ they are $-2/3$ and $-4/3$, illustrating the sign theorem without using a divergent series. The poles at $s = 0$ reflect a singular primitive basis, not a failure of the underlying distribution relations.

5.2 Denominator certificates and construction cost

Let M_p also denote, in this paragraph only, the pushforward matrix on grid vectors defined by

$$(M_p f)_r = \sum_{pu \equiv r \pmod{q}} f_u.$$

To avoid confusion with the integer complement in Section 4, write that complement as $q_p = q/p^e$ here and put $h_p = \text{ord}_{q_p}(p)$. Multiplication of residues gives $M_p^{e+h_p} = M_p^e$. For $z = p^{-s}$,

$$(I - zM_p)^{-1} = \sum_{j=0}^{e-1} z^j M_p^j + \frac{z^e}{1 - z^{h_p}} \sum_{j=0}^{h_p-1} z^j M_p^{e+j}. \quad (5.9)$$

This follows first from a geometric series for $|z| < 1$, then as a rational matrix identity. The commuting product of these inverses, applied to vectors supported on U_q , is the extension $C(s)$.

Proposition 5.6 (A uniform rational denominator). *For integer $k \geq 1$, every reduced denominator of $C_{r,u}(k)$ divides*

$$D_q(k) = \prod_{p^e \parallel q} p^{k(e-1)} (p^{kh_p} - 1). \quad (5.10)$$

Its bit length is $O(kq \log q)$. The coefficients of the raw derivative $C_{r,u}^{(j)}(k)$, viewed as a polynomial in the symbols $\log p$, have denominators dividing $D_q(k)^{j+1}$.

Proof. Every scalar coefficient on the right of (5.9) has denominator dividing $p^{k(e-1)}(p^{kh_p} - 1)$. Multiplication over primes proves the first claim. Since $h_p \leq q$ and $\sum_{p|q} \log p \leq \log q$, its logarithm is at most a constant times $kq \log q$. The matrix product can be written as a ratio of integer polynomials in p^s , with denominator obtained from (5.10) by replacing k with s . Repeated differentiation introduces polynomial factors in $\log p$ and raises the denominator power by at most one each time. For normalized Taylor coefficients, an additional factor $j!$ may be needed. $\square$

The implementation computes (5.3) by convolution on residue classes modulo m , separately for each $m \mid q$. At a fixed m , each prime update takes at most $\varphi(m)^2$ rational arithmetic operations. A complete matrix can therefore be constructed in

$$O\left(\omega(q) \sum_{m|q} \varphi(m)^2 + q\varphi(q)\right) = O(\omega(q)q^2)$$

rational arithmetic operations, using $\sum_{m|q} \varphi(m) = q$. This is an arithmetic-operation bound, not a constant-cost bit model or a claim of optimality. The denominator estimate supplies a conservative size control at positive integer orders.

6 All-orders Hurwitz–Stieltjes identities

Write the Laurent expansion as

$$\zeta(1+t, x) = \frac{1}{t} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(x) t^n, \quad x > 0. \quad (6.1)$$

Here $\gamma_n^{(k)}$ always means k derivatives in x ; a derivative on $C(s)$ is in the spectral variable. The coefficient identities in this section are analytic theorems, distinct from formal rank claims.

Theorem 6.1 (Derived Stieltjes reduction). *For $k \geq 1$, $n \geq 0$, and $x_r = r/q$ with the endpoint convention,*

$$\gamma_n^{(k)}(x_r) = \sum_{u \in U_q} \sum_{j=0}^n \binom{n}{j} W_{r,u}^{[j]}(k+1) \gamma_{n-j}^{(k)}(u/q). \quad (6.2)$$

All the coefficient functions are given by the finite formula (5.3) and its derivatives.

Proof. Differentiating Hurwitz zeta in its parameter yields

$$\partial_x^k \zeta(1+t, x) = (-1)^k (1+t)_k \zeta(k+1+t, x).$$

The pole in (6.1) is independent of x , so for $k \geq 1$ the left side is $\sum_n (-1)^n \gamma_n^{(k)}(x) t^n / n!$. Apply (5.2) at spectral parameter $k+1+t$ and multiply by the common scalar $(-1)^k (1+t)_k$. Taylor multiplication gives (6.2), because $(-1)^j C_{r,u}^{(j)} = W_{r,u}^{[j]}$. $\square$

This proof also explains why the harmonic-number master identity in Chapter 8 is compatible with the same reduction matrix at every Stieltjes order: the rising factorial is common to every rational argument. It does not create new independent distribution coordinates.

Theorem 6.2 (Underived Stieltjes reduction, with the pole term). *For $n \geq 0$,*

$$\gamma_n(x_r) = \sum_{u \in U_q} \sum_{j=0}^n \binom{n}{j} W_{r,u}^{[j]}(1) \gamma_{n-j}(u/q) - \frac{1}{n+1} \sum_{u \in U_q} W_{r,u}^{[n+1]}(1). \quad (6.3)$$

Proof. Apply (5.2) at $s = 1+t$. The coefficients are analytic there. Multiplying their Taylor expansions by (6.1), the coefficient of t^n receives the usual product terms and also $\sum_u C_{r,u}^{(n+1)}(1)/(n+1)!$ from the pole $1/t$. Multiplication by $(-1)^n n!$ converts that additional term to the negative last term of (6.3). The coefficient of $1/t$ agrees because $\sum_u C_{r,u}(1) = 1$. $\square$

Omitting the last term would already give an incorrect formula for γ_0 . The point is not a rank correction: a known scalar right side does not change a homogeneous rank. It is a correction to the actual analytic value identity.

Example 6.3 (An explicit first-order identity). From (5.7), direct differentiation at $s = 1$ gives

$$W_{2,1}^{[1]}(1) = \frac{10}{9} \log 2, \quad W_{2,5}^{[1]}(1) = \frac{8}{9} \log 2.$$

Consequently

$$\gamma_0(1/3) = \frac{2}{3} \gamma_0(1/6) + \frac{1}{3} \gamma_0(5/6) - 2 \log 2. \quad (6.4)$$

At $s = 2$, the analogous coefficients give

$$\begin{aligned} \gamma_1'(1/3) &= \frac{4}{15} \gamma_1'(1/6) + \frac{1}{15} \gamma_1'(5/6) \\ &\quad + \frac{68 \log 2}{225} \gamma_0'(1/6) + \frac{32 \log 2}{225} \gamma_0'(5/6). \end{aligned} \quad (6.5)$$

Since $\gamma_0' = -\psi'$, the signs can also be checked against the parameter-differentiated digamma multiplication identity. The verification program checks (6.3) through $n = 2$ and (6.5) by numerical parameter differentiation of Stieltjes functions.

7 Cyclotomic polylogarithm traces

The same finite module applies to a second evaluation. Put $\zeta_m = \exp(2\pi i/m)$ and evaluate $e_x \mapsto \text{Li}_s(e^{2\pi i x})$. For $\text{Re } s > 1$, grouping the defining series by multiples of p gives

$$\sum_{py=x} \text{Li}_s(e^{2\pi i y}) = p^{1-s} \text{Li}_s(e^{2\pi i x}).$$

Thus the weights are now $A_p = p^{1-s}$, and analytic continuation extends the identity wherever the individual expressions are defined.

For a primitive character χ of conductor $f \mid q$, define

$$\mathcal{P}_{q,\chi}(s) = \sum_{u \in U_q} \chi(u) \text{Li}_s(\zeta_q^u), \quad \tau(\chi) = \sum_{a \in U_f} \chi(a) \zeta_f^a. \quad (7.1)$$

At conductor one take $\tau(\mathbf{1}) = 1$ and $L(s, \mathbf{1}) = \zeta(s)$.

Theorem 7.1 (Conductor-lifted trace identity). *As a meromorphic identity, and with removable singularities interpreted by continuation,*

$$\boxed{\mathcal{P}_{q,\chi}(s) = (q/f)^{1-s} \prod_{\substack{p \mid q \\ p \nmid f}} (1 - \chi(p)p^{s-1}) \tau(\chi) L(s, \bar{\chi}).} \quad (7.2)$$

For every $q > 1$, the left side is an entire function of the spectral variable s .

Proof. At primitive conductor $f > 1$, absolute convergence and the primitive Gauss-sum identity give

$$\sum_{a \in U_f} \chi(a) \text{Li}_s(\zeta_f^a) = \sum_{n \geq 1} \frac{\sum_a \chi(a) \zeta_f^{an}}{n^s} = \tau(\chi) L(s, \bar{\chi}).$$

The Gauss-sum convention, including the complex conjugate on χ , is the one in [6]. Applying (2.5) with $A_p = p^{1-s}$ multiplies this primitive trace by $E_{q,\chi}$. Extracting p^{1-s} from each first-appearance factor gives precisely the factor in (7.2). The conductor-one case follows directly from $\text{Li}_s(1) = \zeta(s)$ and the same polynomial identity.

For completeness, when $z = \zeta_q^u \neq 1$, regrouping by residues gives

$$\text{Li}_s(z) = q^{-s} \sum_{a=1}^q z^a \zeta(s, a/q). \quad (7.3)$$

Hurwitz zeta has only its simple pole at $s = 1$, whose coefficient in this finite sum is $\sum_a z^a = 0$. Hence $\text{Li}_s(z)$ is entire in s . All arguments in the primitive level- q trace differ from one. $\square$

Remark 7.2 (The character phase is consequential). For the order-four character modulo five with $\chi(2) = i$, the level-ten lifting factor is $2^{1-s} - i$, not $2^{1-s} + i$. A complex-character numerical check is included to catch an accidental interchange of χ and $\bar{\chi}$. Real quadratic examples alone would not detect that error.

7.1 Exact nonprincipal vanishing orders

For a nonprincipal primitive χ , set

$$\rho_\chi = \#\{p \mid q : p \nmid f, \chi(p) = 1\}.$$

These are the same conductor statistics that controlled the primitive-grid jet defects at Hurwitz order zero.

Theorem 7.3 (Character trace zeros at order one). *The entire function $\mathcal{P}_{q,\chi}(s)$ has a zero of exact order ρ_χ at $s = 1$, where order zero means a nonzero value. All lower derivatives vanish, and*

$$\begin{aligned} \mathcal{P}_{q,\chi}^{(\rho_\chi)}(1) &= (-1)^{\rho_\chi} \rho_\chi! \tau(\chi) L(1, \bar{\chi}) \\ &\quad \times \prod_{\substack{p \mid q, p \nmid f \\ \chi(p)=1}} \log p \prod_{\substack{p \mid q, p \nmid f \\ \chi(p) \neq 1}} (1 - \chi(p)). \end{aligned} \quad (7.4)$$

Proof. Each factor with $\chi(p) = 1$ in (7.2) is $1 - p^{s-1} = -(s-1) \log p + O((s-1)^2)$. All other displayed factors are nonzero at $s = 1$. A primitive Gauss sum is nonzero, and Dirichlet's nonvanishing theorem states $L(1, \bar{\chi}) \neq 0$ for nonprincipal characters [6, Eq. 25.15.9]. Thus there is no additional zero. Taking the first nonzero Taylor coefficient proves the formula. $\square$

Increasing the exponents of primes already present in q changes higher Taylor coefficients but not this first nonzero derivative: $(q/f)^{1-s}$ equals one at $s = 1$. This invariance provides an additional family of identities between traces at different levels.

7.2 The principal trace and its pole cancellation

Write $\mathcal{P}_q = \mathcal{P}_{q,1}$ and define

$$r = \omega(q), \quad P = \prod_{p \mid q} \log p, \quad B = \log q - \frac{1}{2} \log \text{rad}(q).$$

The conductor-one factor contains a pole, so the vanishing order is one less than the number of primes, not equal to that number.

Theorem 7.4 (Principal trace jet generator). *In a neighborhood of $t = 0$,*

$$\boxed{\mathcal{P}_q(1+t) = (-1)^r P t^{r-1} e^{-Bt} [t\zeta(1+t)] \prod_{p \mid q} \frac{\sinh(t \log p/2)}{t \log p/2}.} \quad (7.5)$$

In particular its zero at $s = 1$ has exact order $r - 1$, and

$$\mathcal{P}_q^{(j)}(1) = 0 \quad (0 \leq j < r - 1), \quad (7.6)$$

$$\mathcal{P}_q^{(r-1)}(1) = (-1)^r (r-1)! P, \quad (7.7)$$

$$\mathcal{P}_q^{(r)}(1) = (-1)^r r! P(\gamma - B), \quad (7.8)$$

$$\mathcal{P}_q^{(r+1)}(1) = (-1)^r (r+1)! P \left[-\gamma_1 - B\gamma + \frac{B^2}{2} + \frac{1}{24} \sum_{p \mid q} (\log p)^2 \right]. \quad (7.9)$$

Here γ_1 is the ordinary first Stieltjes constant in (6.1).

Proof. The principal case of (7.2) is

$$\mathcal{P}_q(1+t) = q^{-t} \prod_{p|q} (1-p^t) \zeta(1+t).$$

Use $1 - e^x = -xe^{x/2} \sinh(x/2)/(x/2)$ in every factor. This proves (7.5). The factors in square brackets and in the hyperbolic-sine product are analytic and equal to one at zero. Expanding them gives

$$t\zeta(1+t) = 1 + \gamma t - \gamma_1 t^2 + O(t^3), \quad \prod_p \frac{\sinh(t \log p/2)}{t \log p/2} = 1 + \frac{t^2}{24} \sum_p (\log p)^2 + O(t^4).$$

Multiplication by e^{-Bt} gives all the displayed derivatives. $\square$

7.3 Concrete exact identities

The following examples are proved consequences of the preceding theorems, not integer-relation conjectures. The derivatives are in the *order* of the polylogarithm, not its argument.

At level thirty,

$$\sum_{u \in U_{30}} \text{Li}_1(\zeta_{30}^u) = 0, \tag{7.10}$$

$$\sum_{u \in U_{30}} \partial_s^1 \text{Li}_s(\zeta_{30}^u) \Big|_{s=1} = 0, \tag{7.11}$$

$$\sum_{u \in U_{30}} \partial_s^2 \text{Li}_s(\zeta_{30}^u) \Big|_{s=1} = \boxed{-2 \log 2 \log 3 \log 5}. \tag{7.12}$$

The third derivative of this trace is $-6 \log 2 \log 3 \log 5 [\gamma - \frac{1}{2} \log 30]$. At level 210, derivatives of orders zero, one, and two vanish, while

$$\sum_{u \in U_{210}} \partial_s^3 \text{Li}_s(\zeta_{210}^u) \Big|_{s=1} = 6 \log 2 \log 3 \log 5 \log 7. \tag{7.13}$$

For the quadratic character χ_3 with $\chi_3(1) = 1$ and $\chi_3(2) = -1$, the primitive trace at order one is $\tau(\chi_3)L(1, \chi_3) = i\pi/3$. Since $\chi_3(7) = \chi_3(13) = 1$, one obtains

$$\sum_{u \in U_{21}} \chi_3(u) \partial_s^1 \text{Li}_s(\zeta_{21}^u) \Big|_{s=1} = -\frac{i\pi}{3} \log 7, \tag{7.14}$$

$$\sum_{u \in U_{273}} \chi_3(u) \partial_s^2 \text{Li}_s(\zeta_{273}^u) \Big|_{s=1} = \frac{2i\pi}{3} \log 7 \log 13. \tag{7.15}$$

In the first equation the order-zero trace vanishes; in the second, both order-zero and first-order traces vanish. Likewise the character χ_4 gives

$$\sum_{u \in U_{20}} \chi_4(u) \partial_s^1 \text{Li}_s(\zeta_{20}^u) \Big|_{s=1} = -\frac{i\pi}{2} \log 5. \tag{7.16}$$

These primitive trace constants follow directly from $\text{Li}_1(z) = -\log(1-z)$ with the values continued from inside the unit disk, so the signs do not depend on an unspecified logarithm branch.

For the real quadratic character χ_5 , positive on residues 1, 4 and negative on 2, 3, let $\phi = (1 + \sqrt{5})/2$. Pairing conjugates in the primitive trace gives

$$\mathcal{P}_{5,\chi_5}(1) = 2 \log \frac{\sin(2\pi/5)}{\sin(\pi/5)} = 2 \log \phi.$$

As $\chi_5(11) = 1$,

$$\sum_{u \in U_{55}} \chi_5(u) \left. \partial_s^1 \text{Li}_s(\zeta_{55}^u) \right|_{s=1} = -2 \log \phi \log 11. \quad (7.17)$$

This derivation needs no numerical integer-relation search or unproved independence statement about logarithms.

7.4 A stable independent formula for numerical checks

Differentiation of (7.3) at its cancelled pole gives, for $z = \zeta_q^u \neq 1$,

$$\left. \partial_s^k \text{Li}_s(z) \right|_{s=1} = \frac{(-1)^k}{q} \sum_{a=1}^q z^a \sum_{j=0}^k \binom{k}{j} (\log q)^{k-j} \gamma_j(a/q). \quad (7.18)$$

There is no pole term, because its complete Fourier coefficient vanishes identically. This supplies a finite Stieltjes-coordinate evaluator for the trace identities, independently of their Euler-factor right sides. In development, direct tiny-step numerical differentiation of a polylogarithm routine near integral order gave an unreliable trace derivative; the pole-cancelled formula avoids that cancellation problem. This observation is an implementation diagnostic, not a claimed theorem about all versions of a library.

8 The Gaussian S_4 conjecture: an exact obstruction

The manuscript uses decreasing nested indices:

$$\text{Li}_{a,b}(x, y) = \sum_{m>n \geq 1} \frac{x^m y^n}{m^a n^b}, \quad g_{a,b} = \text{Im Li}_{a,b}(i, 1), \quad G = \beta(2).$$

Its equation `gauss:eq:S4-closed` is the following candidate.

Conjecture 8.1 (The manuscript's weight-five evaluation). With $H_0 = 0$ and $H_n = \sum_{j=1}^n 1/j$,

$$S_4 := \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^4} = \frac{4g_{4,1} - 3g_{3,2} - 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27G\zeta(3)}{224} - 2\beta(4) \log 2. \quad (8.1)$$

We retain this as a conjecture here. The new rigorous result is a specific obstruction to deriving it from products of two single polylogarithms alone, in the exact presentation defined next. This is more informative than a failed numerical search, but it must not be misread as an obstruction to all possible proofs.

Lemma 8.2 (An exact even-inner-index filter). *One has*

$$S_4 = \text{Im}(\text{Li}_{4,1}(i, 1) + \text{Li}_{4,1}(i, -1)). \quad (8.2)$$

Proof. The inner sum in the right side is

$$\sum_{n < m} \frac{1 + (-1)^n}{n} = H_{\lfloor (m-1)/2 \rfloor}.$$

The imaginary part of i^m vanishes at even m and is $(-1)^j$ at $m = 2j + 1$. Substitution gives the defining sum for S_4 . The double sums are absolutely convergent, so this rearrangement is legitimate. $\square$

8.1 A fully specified finite law vocabulary

Introduce formal coordinates

$$X_{a;r,t} = \text{Im Li}_{a,5-a}(i^r, i^t), \quad 1 \leq a \leq 4.$$

Complex conjugation identifies $X_{a;-r,-t} = -X_{a;r,t}$, and a pair of even residues has imaginary part zero. Choose the six representatives

$$(r, t) \in \{(0, 1), (1, 0), (1, 1), (1, 2), (1, 3), (2, 1)\}. \quad (8.3)$$

They give 24 double coordinates. Append three formal single-value product coordinates

$$A = \pi^5, \quad B = G\zeta(3), \quad C = \beta(4) \log 2.$$

For each $p + v = 5$, $p, v \geq 1$, and $r, t \in \mathbb{Z}/4\mathbb{Z}$, include the imaginary parts of the two identities

$$\text{Li}_p(i^r) \text{Li}_v(i^t) = \text{Li}_{p,v}(i^r, i^t) + \text{Li}_{v,p}(i^t, i^r) + \text{Li}_5(i^{r+t}), \quad (8.4)$$

$$\begin{aligned} \text{Li}_p(i^r) \text{Li}_v(i^t) &= \sum_{j=0}^{p-1} \binom{v-1+j}{j} \text{Li}_{v+j,p-j}(i^t, i^{r-t}) \\ &\quad + \sum_{j=0}^{v-1} \binom{p-1+j}{j} \text{Li}_{p+j,v-j}(i^r, i^{t-r}). \end{aligned} \quad (8.5)$$

The first is the colored stuffle law. The second follows by shuffling $0^{p-1}(i^r)^{-1}$ with $0^{v-1}(i^t)^{-1}$ in the outer-letter-first iterated-integral convention; the binomial coefficients count the interspersed zero letters.

Use the standard single-value reductions at fourth roots of unity. Their imaginary products at total weight five lie in the three atoms A, B, C . At the divergent factor $\text{Li}_1(1)$, use regularized constant term zero. Only a linear divergence polynomial occurs here: a weight-one factor can multiply a convergent weight-four factor, but two divergent weight-one factors cannot occur at total weight five in this vocabulary. Thus shuffle and stuffle regularization agree on the degree-zero and degree-one terms being used; the general regularization framework is standard [7]. The formal row-space result below would remain well-defined even if these boundary rows were omitted; we report that smaller matrix separately.

After conjugation reduction and removal of zero rows there are 96 rows and 27 columns. Duplicate nonzero rows are intentionally retained, so the law-generation specification is simple and replayable. Without products containing $\text{Li}_1(1)$ there are 88 rows.

Theorem 8.3 (A finite row-space obstruction). *Let M be the 96×27 rational matrix just defined. Its rank is 20. The row corresponding, via (8.2), to 224 times the difference between the two sides of (8.1) is*

$$T = 96X_{4;1,0} + 96X_{3;1,0} + 288X_{2;1,0} + 224X_{4;1,2} - A + 27B + 448C. \quad (8.6)$$

It does not lie in the row span of M . The augmented matrix has rank 21. The unregularized 88-row matrix has rank 19.

Proof. The nonmembership assertion has a particularly small certificate. Define a linear functional w on the 27 coordinates by the following eight nonzero entries, and set it to zero on every other coordinate, including A, B, C :

Coordinate	w	Coordinate	w
$X_{1;1,0}$	-2	$X_{2;0,1}$	1
$X_{2;1,0}$	3	$X_{2;1,3}$	-1
$X_{3;0,1}$	-3	$X_{3;1,0}$	-1
$X_{3;1,3}$	-1	$X_{4;0,1}$	2

Substitution in the two finite row formulas (8.4)–(8.5), for their stated index ranges, gives $Mw = 0$. This is 96 exact rational zero checks, recorded in the certificate. On the target, however,

$$w(T) = 96(-1) + 288(3) = 768 \neq 0.$$

Thus no linear combination of the declared rows equals T . The rank assertions are additional finite exact calculations: Gaussian elimination gives ranks 20, 21, 19, respectively. Both the generator using symbolic rational arithmetic and an independent standard-library rational replay program are supplied. The nonmembership proof itself needs only the displayed eight-entry witness, not the rank calculation. $\square$

This theorem remains true after adding arbitrary relations supported only on the three product atoms, because w annihilates those atoms. It does not remain a theorem about an enlarged multiple-polylogarithm presentation unless the new rows are also checked. Products involving doubles, elimination through depth three, distribution at other levels, or functional equations at transformed arguments may provide the missing relation. The witness directs future searches toward precisely the relations on which it does not vanish.

8.2 Fresh numerical evidence, with its exact limitation

An independent integral comparison uses

$$S_4 = -\frac{1}{6} \int_0^1 \frac{(-\log x)^3 \log(1+x^2)}{1+x^2} dx, \quad (8.7)$$

$$\text{Li}_{a,b}(z, 1) = \frac{1}{(a-1)!} \int_0^1 (-\log t)^{a-1} \frac{z \text{Li}_b(zt)}{1-zt} dt. \quad (8.8)$$

For (8.7), expand $\sum_{n \geq 0} H_n(-x^2)^n = -\log(1+x^2)/(1+x^2)$ and integrate. For (8.8), interchange absolutely convergent power series inside the integral and use $\int_0^1 t^{m-1}(-\log t)^{a-1} dt = (a-1)!/m^a$. The relevant boundary values follow by dominated convergence.

At 55 decimal working digits the independent integral evaluations give

$$S_4 = -0.01049345629733504345335675100620175196351061887731976151 \dots$$

and the residual in (8.1) is approximately 7.17×10^{-57} . This is floating-point agreement, not an interval proof and not a proof that the residual is exactly zero. The claim that has been proved is the finite-presentation obstruction, not the conjectured evaluation.

9 Corrections and clarifications for the manuscript

9.1 Replace the distribution-rank experiment by a defined theorem

Replace the experimental-only formulation at `tower:thm:rank` by Corollary 3.1, with an explicit prime/divisor row definition. Retain the explanation that formal quotient dimension does not prove numerical independence. The same addition supplies the uniform parity count in Chapter 7 and all-order jet extensions in Chapter 8. The historical data may remain as provenance, but should not be described as the proof of the uniform statement.

At exceptional weights, do not infer a rank jump from a failed inverse. Theorem 2.2 gives a globally valid conductor basis; Theorems 4.1 and 4.3 identify the separate failure of primitive-grid coordinates. A proof based only on their inversion would miss this distinction.

9.2 The cubic log-gamma moment already has a named-constant formula

Chapter 7, `integral:neg:cubic`, correctly observes that a small basket of single zeta derivatives did not explain the cubic moment. However, its claim that finding the appropriate named Tornheim-type atoms is still open is superseded by Bailey–Borwein–Borwein: their paper explicitly evaluates the cubic moment in Theorem 6 and equation (11), using derivatives of Tornheim–Witten zeta functions [8]. The appropriate correction is to distinguish an existing formula in that larger vocabulary from the separate question of reduction to a smaller vocabulary. Appendix A gives the formula in the notation used here and an independent Fourier coefficient derivation.

This is a literature correction, not a new evaluation claimed by this article. It does not imply that all derivatives appearing there reduce to ordinary zeta derivatives or classical single polylogarithms.

9.3 At mixed Gaussian points the relevant involution swaps arguments

The stuffle relation identifies a symmetric combination under

$$(a, b; x, y) \mapsto (b, a; y, x),$$

not merely $(a, b) \mapsto (b, a)$ with (x, y) fixed. Thus at a mixed point the complementary antisymmetric combination is

$$\text{Li}_{a,b}(x, y) - \text{Li}_{b,a}(y, x).$$

It need not vanish when $a = b$ and $x \neq y$. The manuscript’s “all depth-two content” sentence following its symmetric Gaussian formulas should be restricted to the paired involution. Nothing in this correction asserts that a nonzero antisymmetric combination is numerically independent of single-value products.

Proposition 9.1 (A nonzero mixed diagonal). *One has the strict inequality*

$$\text{Im}(\text{Li}_{2,2}(-1, i) - \text{Li}_{2,2}(i, -1)) > 0. \quad (9.1)$$

Proof. The general double-integral formula, with $\text{Li}_b(xyt)$ in the numerator, gives their difference as

$$- \int_0^1 (-\log t) \frac{(1+i) \text{Li}_2(-it)}{(1+t)(1-it)} dt.$$

Write $\text{Ti}_2(t) = \int_0^t \arctan(v) dv/v > 0$ for $0 < t \leq 1$. Then $\text{Li}_2(-it) = \frac{1}{4} \text{Li}_2(-t^2) - i \text{Ti}_2(t)$. The imaginary part of the displayed integrand, including $-\log t$, is

$$\frac{-\log t}{(1+t)(1+t^2)} \left[(1-t) \text{Ti}_2(t) - \frac{1+t}{4} \text{Li}_2(-t^2) \right].$$

It is strictly positive on $(0, 1)$, since $\text{Li}_2(-t^2) < 0$. The integral converges, proving (9.1). $\square$

9.4 Keep three meanings of a certificate distinct

The polynomial row certificates prove formal identities from a defined law vocabulary. Their analytic evaluations require the convergence and continuation arguments supplied above. Floating-point residuals test implementations but prove neither formal row membership nor equality of general transcendental numbers. In particular, Theorem 8.3 and the small numerical residual for S_4 are fully compatible facts. A proof-assistant port would add another layer of verification; none is claimed for this package.

10 Reproducible computations and integration

10.1 Exact checks actually executed

The package’s exact verification program tests every $2 \leq q \leq 60$. For each denominator, it uses five weight patterns: $A_p = p^2$, all $A_p = 1$, all $A_p = 0$, all $A_p = -1$, and a mixed zero/nonzero pattern. In each of the resulting 295 cases it checks the full rank, anchored rank, and applicable even/odd ranks with exact rational arithmetic. These include deliberately resonant and zero weights.

A second set of 295 cases uses spectral orders $s = -2, -1, 1, 2, 3$ for the same denominators. It constructs (5.3) exactly and checks every divisor row, the identity primitive submatrix, the predicted support counts, the sign rule, and the endpoint formula. Negative orders use finite rational continuation, not an infinite series. Four complete matrices are included as JSON certificates, for $(q, s) = (12, 2), (21, -1), (30, 3), (60, 2)$.

The character verification covers the same denominator range. Characters are represented by rational phases modulo one, not approximate complex numbers. It checks conductor counts, the root multiplicities and pure powers in (4.2), and the central primitive-jet rank profiles. Character examples for levels 12, 21, 30 are supplied.

The S_4 matrix and the eight-entry obstruction witness are regenerated from the declared laws, then replayed independently with `fractions.Fraction`. The replay checks all 96 annihilations, the target pairing 768, and ranks 20 and 21. Finite regression ranges support the software; the uniform statements are established by the proofs, not by extrapolation from these tests.

10.2 Numerical diagnostics actually executed

The numerical program was run at 55 decimal working digits. It checks seven trace/phase identities, three underived Stieltjes identities, one parameter-differentiated Stieltjes identity, and the S_4 integral comparison. Trace derivatives use (7.18) with additional guard digits, while a nonintegral-order complex-character phase check uses direct polylogarithms. No target Euler-factor identity is used to evaluate its own trace side.

Diagnostic family	Largest observed absolute residual
Trace and complex-character checks	5.77×10^{-56}
Underived Stieltjes checks, $n = 0, 1, 2$	8.16×10^{-56}
Derived Stieltjes check, $k = n = 1$	3.27×10^{-55}
S_4 independent integral comparison	7.17×10^{-57}

Residuals smaller than the working precision can occur because rounded real parts coincide or because extra guard digits are retained. They are not claims of that many correct decimal places. These experiments use arbitrary-precision floating point, not interval arithmetic, and the JSON records say so explicitly.

10.3 Files and reproduction

The article is self-contained in `article/distribution_jets.tex`. The principal programs are:

File under code/	Purpose
<code>distribution.py</code>	Exact distribution rows, rational ranks, finite normal forms.
<code>verify_exact.py</code>	Uniform-range regression checks and matrix certificates.
<code>verify_characters.py</code>	Exact character phases, determinant factors, and jet profiles.
<code>verify_s4_presentation.py</code>	Generate the declared Gaussian row system and witness.
<code>replay_certificate.py</code>	Standard-library replay of the S_4 certificate.
<code>verify_numeric.py</code>	Separate trace, Stieltjes, and S_4 numerical diagnostics.

The distribution and replay modules use only the Python standard library. The character generator and symbolic Gaussian generator use SymPy; numerical diagnostics use mpmath. The versions used in this run are recorded in the package. The `integration/` directory provides editorial replacement text and a label map rather than silently rewriting the live manuscript. Source paths, inspected Git blob hashes, and the repository commit are recorded in `provenance.json`. No repository file was changed remotely.

11 Further research questions

11.1 Complete the proof of the weight-five conjecture

The most direct next problem is to prove or refute Conjecture 8.1. The exact witness makes an enlarged search more targeted: generate candidate new laws, apply w first, and prioritize laws with nonzero pairing. Promising sources include products of a single and a double polylogarithm with elimination of triples, regularized distribution relations, and argument-changing functional equations. A successful proof should include the additional law instances and an exact final row combination, not just a larger integer-relation search.

11.2 Integral and modular versions of the weighted module

The proof of Theorem 2.2 divides by character group orders and works over a characteristic-zero splitting field. What is the exact integral structure of the independently weighted module? At ordinary weights, classical integral freeness provides an important benchmark. At general weights, one should determine torsion and the primes at which the conductor Fourier method ceases to describe the integral lattice. In positive characteristic, nonsemisimple character actions can change the argument; the characteristic-zero theorem alone does not settle that problem.

11.3 Multivariate jets and intersection multiplicities

Independent variables t_p in $A_p = \exp(t_p)$ give a multivariate version of the normal form. The module remains free by base change, but the primitive-grid inclusion has a union of coordinate and cyclotomic divisors rather than one-variable Smith factors. It would be useful to compute its Fitting

ideals, truncated multigraded ranks, and optimal minimal jet data. The determinant gives the first invariant, not the entire multivariate classification.

11.4 Sparse algorithms and certified numerical evaluation

The residue-subgroup support theorem can be used to avoid dense matrices. Can one give output-sensitive complexity bounds in terms of $\sum_r |\text{supp } C_r|$, rather than q^2 ? Can the conductor basis be combined with interval arithmetic to evaluate spectral jets uniformly near resonances, without catastrophic cancellation? The coefficients have explicit denominators at positive integer orders; a certified remainder analysis for analytic jets would complement that exact arithmetic structure.

11.5 Trace identities at greater depth

For multiple polylogarithms, distributing a root argument interacts with several nested indices and with prefix products. A useful extension would identify an explicit finite module whose degree-one case is the present theory and whose higher-depth relations remain compatible with shuffle and stuffle. One should not assume that the rank simply becomes a power of $\varphi(q)$: depth, convergence, regularization, and diagonal terms all enter.

11.6 Arithmetic meaning of the trace-jet constants

Theorems 7.3 and 7.4 identify exact vanishing orders and first nonzero coefficients. The higher coefficients involve L -derivatives, Stieltjes constants, and logarithms. Which additional functional equations reduce their arithmetic vocabulary? Any claim of minimality must specify a comparison algebra and distinguish formal, motivic, and numerical independence. The present rank theorem alone gives none of those numerical lower bounds.

11.7 Log-gamma moments beyond the known Tornheim formula

The cubic moment should be studied starting from its existing Tornheim–Witten derivative expression, not by rediscovering the named atoms. Useful concrete questions include reductions among the particular derivatives T_c, T_{ac}, T_{ab} , better certified evaluation of those derivatives, and the organization of higher log-gamma moments by Fourier resonance patterns. Any proposed smaller closed form should be tested against the established larger-vocabulary identity.

11.8 A proof-assistant development

A staged formalization can begin with the finite module: define the fiber rows, prove prime sufficiency, establish character-block edges, and verify the monic conductor normal form. This isolates finite algebra from analytic continuation. The second stage can formalize Hurwitz and polylogarithm evaluations and their local coefficient identities. The small S_4 annihilator is a separate attractive target for a finite rational certificate checker. None of these future formalizations is asserted to have been completed by the present work.

A The known cubic log-gamma formula in Tornheim notation

Define, in an open neighborhood of $(1, 1, 1)$,

$$T(a, b, c) = \sum_{m, n \geq 1} \frac{1}{m^a n^b (m+n)^c}.$$

The series and all finite partial derivatives there converge absolutely. Let T_c, T_{ac}, T_{ab} mean the indicated partial derivatives evaluated at $(1, 1, 1)$, and put $L = \log(2\pi)$, $A = \gamma + L$. A form of the Bailey–Borwein–Borwein evaluation [8, Eq. (11), Theorem 6] is

$$\begin{aligned} \int_0^1 \log^3 \Gamma(x) dx &= \frac{L^3}{8} + \frac{L\pi^2}{32} + \frac{LA^2}{8} - \frac{3LA\zeta'(2)}{2\pi^2} + \frac{3L\zeta''(2)}{4\pi^2} + \frac{3\zeta(3)}{16} \\ &+ \frac{3}{8\pi^2} [2A^2\zeta(3) - 2AT_c + 2T_{ac} - T_{ab}]. \end{aligned} \quad (\text{A.1})$$

Here $T_c = 2\partial_u \zeta(u, 1)|_{u=2}$, where in this sentence $\zeta(u, 1) = \sum_{m>n \geq 1} m^{-u} n^{-1}$ is the *double zeta*, not the Hurwitz zeta. This follows by partial fractions and identifies our notation with the published formula.

A direct derivation is short. Kummer's Fourier expansion gives the mean $L/2$ and positive-frequency coefficients

$$\alpha_n = \frac{1}{4n} - \frac{i(A + \log n)}{2\pi n} \quad (n \geq 1)$$

for $\log \Gamma(x)$. The negative-frequency coefficients are their complex conjugates. Taking the zero-frequency coefficient of the cube yields

$$(L/2)^3 + 6(L/2) \sum_{n \geq 1} |\alpha_n|^2 + 6 \operatorname{Re} \sum_{m, n \geq 1} \alpha_m \alpha_n \overline{\alpha_{m+n}}. \quad (\text{A.2})$$

The first sum is

$$\sum_{n \geq 1} |\alpha_n|^2 = \frac{\pi^2}{96} + \frac{A^2}{24} - \frac{A\zeta'(2)}{2\pi^2} + \frac{\zeta''(2)}{4\pi^2}.$$

For the double sum, expand the three factors. The constant term uses $T(1, 1, 1) = 2\zeta(3)$. Terms linear in the logarithms leave only the $\log(m+n)$ sum, namely $-T_c$; the quadratic terms give $2T_{ac} - T_{ab}$. Substitution in (A.2) gives (A.1).

To justify the multiplication, first multiply every Fourier coefficient by $\rho^{|n|}$ with $0 < \rho < 1$. The smoothed Fourier series is absolutely convergent. Since $\log \Gamma$ belongs to $L^3(0, 1)$, these Poisson-smoothed functions converge in L^3 , so the integrals of their cubes converge. The logarithmically weighted Tornheim sums are absolutely convergent near $(1, 1, 1)$ and dominate the corresponding smoothed double sums. Passage to the limit in both representations is therefore legitimate. This derivation is included to make the proposed correction mathematically self-contained, not to claim priority for the formula.

B Provenance, claim boundaries, and artifact checks

The research baseline is repository commit

29d9344771b41df39ff9b7890c76b1f152572dde.

The source root is

Analysis/Polylogarithms/docs/manuscript/.

Relevant inspected blob hashes are recorded in the package’s machine-readable provenance file. In particular, the weight-five conjecture was checked against the pinned Chapter 4 blob 6171912ff7724784cabb789a066c61223ab298bc. The article is a companion contribution: it does not claim to audit all eleven chapters or every historical identity store.

Claim	Status in this contribution
Universal weighted rank and conductor normal form	Proved over a characteristic-zero splitting field.
Primitive-grid determinant and local jet ranks	Proved; exact character checks included.
Hurwitz–Stieltjes and cyclotomic trace formulas	Analytic identities proved with stated conventions.
S_4 evaluation	Retained as conjectural; fresh numerical agreement only.
S_4 nonmembership in the declared row span	Proved by a finite exact annihilator.
Mixed Gaussian diagonal antisymmetry	A strictly positive imaginary difference is proved.
Cubic log-gamma evaluation	Existing published result; notation and status corrected.
Numerical independence; minimal transcendental basis	Not claimed.
Proof-assistant verification; external peer review	Not claimed.

The bundled tests separate exact arithmetic from floating-point experiments. Every certificate has a declared meaning and an explicit coordinate convention. A checksum manifest is provided for the package contents. The build and test scripts operate locally and do not write to the remote repository.

References

- [1] V. Reshetnikov, *Polylogarithms and their Arithmetic Bridges*, ProveIt repository, consolidated manuscript, October 2026. Inspected commit 29d9344771b41df39ff9b7890c76b1f152572dde. <https://github.com/VladimirReshetnikov/ProveIt/tree/29d9344771b41df39ff9b7890c76b1f152572dde/Analysis/Polylogarithms/docs/manuscript>.
- [2] Y. Ouyang, *Group cohomology of the universal ordinary distribution*, arXiv:math/9911032 (1999), especially Proposition 3.1. <https://arxiv.org/abs/math/9911032>.
- [3] D. S. Kubert, *The $\mathbb{Z}/2\mathbb{Z}$ cohomology of the universal ordinary distribution*, Bulletin de la Société Mathématique de France **107** (1979), 203–224. https://www.numdam.org/article/BSMF_1979__107__203_0.pdf.
- [4] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. Accessed 9 October 2026. <https://dlmf.nist.gov/25.11>.
- [5] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. Accessed 9 October 2026. <https://dlmf.nist.gov/25.12>.
- [6] NIST Digital Library of Mathematical Functions, §25.15, *Dirichlet L -functions*, including (25.15.4), (25.15.6), and the nonvanishing statement (25.15.9). Accessed 9 October 2026. <https://dlmf.nist.gov/25.15>.
- [7] G. Racinet, *Doubles mélanges des polylogarithmes multiples aux racines de l'unité*, arXiv:math/0202142 (2002). <https://arxiv.org/abs/math/0202142>.
- [8] D. H. Bailey, D. Borwein, and J. M. Borwein, *On Eulerian Log-Gamma Integrals and Tornheim–Witten Zeta Functions*, preprint dated 27 June 2012; Ramanujan Journal **36** (2015), 43–68. DOI: 10.1007/s11139-012-9427-1. <https://carmamaths.org/resources/jon/log-gamma.pdf>.